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Title: real-time texture streaming system for XR applications running on android

Introduction:

Extended Reality(XR) applications often involve large and complex 3D environments with high resolution textures, which can lead to high-resolution textures, which can lead to high memory consumption and performance bottlenecks, especially on resource-constrained devices like Android. Loading all textures at once can result in unnecessary memory usage, impacting rendering performance and causing lag, or crashes.

To address this real time texture streaming allows textures to be dynamically loaded only when needed and unloaded when they are no longer visible. This ensures an optimal balance between visual fidelity and performance, making it possible to render large-scale XR environments efficiently on Android devices.

Objective:

The goal of this project is to design and implement a real-time texture streaming system for XR applications running on android. The system should:

1. **Dynamically load and unload texture**s based on asset visibility in a simulated XR environment.

2. **Optimize memory and GPU usage** by ensuring only required textures are kept in memory.

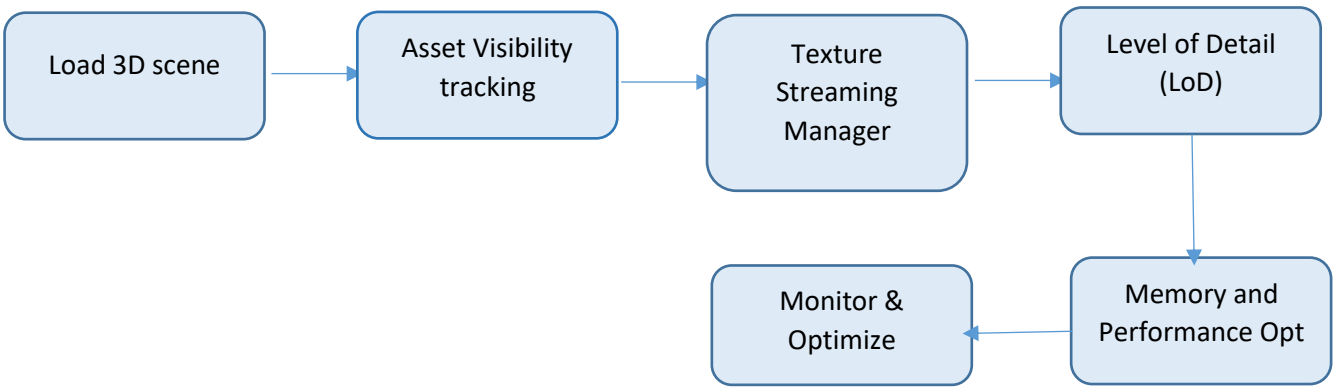
3. **Maintain smooth rendering performance** by efficiently handling texture loading/unloading without causing frame drops.

4. **Ensure scalability** to support large 3D environments with multiple high-resolution assets.

Proposed Solution:

The solution involves:

1- Visibility based Management: Track asset visibility in a virtual 3D scene. Load and unload high-resolution textures only when assets enter and exit out of the field, respectively.
2- Efficient Texture handling Techniques: Implement Level of Detail (LOD) to manage multiple texture resolutions dynamically. Use texture compression to reduce memory footprint.
3- Performance Optimization: Use asynchronous texture loading to prevent frame drops. Minimize CPU-GPU overhead by batch processing texture updates.
4- Testing: Monitor memory consumption, loading times , and rendering performance.



References:

"Adaptive streaming and rendering of textures for large scale virtual environments" by Gosse, E. , Benard, P.(2020). ACM Transactions on Graphics